INCH POUND

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SUPERSEDING
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MILITARY SPECIFICATION

GRINDING MACHINE, CYLINDER HEAD (WET TYPE)

This specification is approved for use by all Department and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers a stationary, motor driven, surface grinding machine for wet grinding cylinder heads, manifolds, and engine blocks for in-line and V-type internal combustion engines and similar flat surfaces.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin
P-D-680	- Dry Cleaning Solvent
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer Resisting
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4910

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PPP-B-1055 - Barrier Material, Waterproof, Flexible
PPP-C-843 - Cushioning Material, Cellulosic
PPP-D-705 - Drum, Shipping and Storage: Steel, 16 and 30 Gallon Capacity
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon (208 Liters)
PPP-P-40 - Packaging and Packing of Hand Tool
PPP-T-60 - Tape, Packaging, Waterproof

MILITARY

MIL-C-104 - Crates, Wood: Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116 - Preservation, Methods of
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-T-152 - Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electronic, and Associated Equipment
MIL-V-173 - Varnish, Moisture-and-Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment)
MIL-P-775 - Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing); and Fittings, Nozzles, and Strainers
MIL-C-3774 - Crates, Wood; Open 12,000- and 16,000- Pound Capacity
MIL-A-3816 - Abrasives and Abrasive Products; Packaging of
MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal
MIL-H-6083 - Hydraulic Fluid, Petroleum Base, For Preservation and Operation
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of
MIL-B-26195 - Boxes, Wood-Cleated, Skidded, Load-Bearing Base
MIL-C-52950 - Crates, Wood, Open and Covered

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-889 - Dissimilar Metals.

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MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGISTER (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B7.1-78 - Safety Requirements for the Use, Care, and Protection of Abrasive Wheels.
ANSI B11.9-75 - Safety Requirements for the Construction, Care, and Use of Grinding Machines.
ANSI B74.2-74 - Specifications for Grinding Wheels, Shapes and Sizes, Mounted Wheels, Sizes and Identification.
ANSI Y14.5M-82 - Dimensioning and Tolerancing.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E380-84 - Standard for Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG-1 - Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street NW, Washington, D.C. 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Metalworking Machine Tools and Plastics Machinery.

(Applications for copies of ISO documents should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one or more grinding machines for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The grinding machine shall be new and of the manufacturer's latest design meeting the performance requirements specified herein. Components of the machine shall be designed to be of the size and strength necessary to sustain the maximum allowable operating load with maximum efficiency and minimum wear. The grinding machine shall be designed with a grinding wheel spindle mounted in the vertical or horizontal position and located either above or below the parts being ground. The grinding wheel

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head, or table, shall provide the traverse and feed motions required. The machine shall be provided with adjustable stops that reverse travel automatically.

3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the grinding machine. In this specification, all measurements, dimensions, sizes and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380-84.

3.2.2 Measuring and indicating devices. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and SI systems of measurement. When a dual US and SI system is furnished, dials shall have independent zero adjustments and shall be calibrated so that the last dial graduation progresses into, and is continuous with, the first dial graduation as the dial is rotated through the zero position. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.3 Maintainability. Design of the grinding machine and the placement of adjusting features shall ensure that operational maintenance capability shall be reduced to the simplest form of procedures, requiring a minimum of skills and common tools. Parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, repair or lubrication without modification, distortion, or damage to the machine or any machine part or component.

3.2.4 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings, and filler caps shall be accessible without requiring disassembly of functional parts. Lubrication reservoirs, if applicable, shall have means for checking levels.

3.2.5 Gears. All gear and pinion designs, used in the grinding machine or its components, shall be manufactured in accordance with the best commercial practices of the machine tool industry. Gears and mountings shall be designed and machined to provide a fit-up to minimize backlash.

3.2.6 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.2.7 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without

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modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.2.8 Safety and health requirements. Safety devices shall be provided for all parts presenting safety hazards. Guards shall be removable to provide easy access to guarded parts. The grinding machine shall comply with the general safety and health requirements promulgated under 29 CFR 1910 (Federal OSHA Standards) that are applicable to the machine itself. In addition, the grinding machine shall comply with those requirements of ANSI B11.9 that are specified therein as the responsibility of the machine manufacturer. In the event of conflict between the OSHA and ANSI standards, the OSHA requirements shall apply. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.4).

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the machine tool industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.4 Construction . The grinding machine shall be constructed to be capable of performing the operations specified herein. The machine shall be complete so that, when installed and connected to a proper power source, the machine may be used for all operations for which it was designed.

3.4.1 Castings and forgings. Castings shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

3.4.2 Welding, brazing, or soldering. Welding, brazing, and soldering shall be of a quality which shall sustain the performance requirements of the welded, brazed, or soldered part. These operations shall not be employed as repair measures for defective parts.

3.4.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. The methods for preventing loss of tightness or adjustment shall be in

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accordance with accepted engineering standards and practice. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

3.4.4 Surfaces. All castings, forgings, molded parts, stampings, and welded parts shall be clean and free from sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

3.4.5 Electrical connections. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. All other electrical connections shall be mechanically secured and soldered or brazed. Only rosin base fluxes shall be used in all soldering operations.

3.5 Performance and product characteristics. The grinding machine shall be capable of wet grinding aluminum, alloy steel, or cast iron cylinder heads, manifolds, and engine blocks of in-line and V-type internal combustion engines and other flat-surfaced parts.

3.5.1 Accuracy. The grinding machine shall be capable of grinding a surface flatness which shall not deviate more than 0.002 inch from the plane of the surface being ground. The surfaces shall parallel the center line of main bearings within 0.002 inch to assure perpendicular alignment of assembled components when measured as defined in ANSI Y14.5M-82.

3.5.2 Capacity. The grinding machine shall have a minimum work capacity not less than 46 inches in length by 14 inches wide. The vertical height capacity of the machine shall be no less than 26 inches. The machine shall provide a transverse motion, between the wheel head and the part, of no less than 120 inches per minute, by moving either the wheel head across the part or the across the wheel head.

3.6 Components. The grinding machine shall consist of, but may not necessarily be limited to, the principal components covered in 3.6.1 through 3.6.6.1.

3.6.1 Base. The grinding machine base shall be constructed of metal and shall be sufficiently rigid to meet the performance characteristics specified in 3.5. Leveling devices and hold-down bolt holes shall be provided.

3.6.2 Table. The table shall be of a sufficient capacity to accommodate the minimum work size specified in 3.5.2 and shall provide support for the work. Necessary provisions shall be furnished for holding the work securely in place for the grinding operation.

3.6.3 Grinding wheel. Grinding wheels for vertical grinding shall be either hollow center cupped or segmented types, not less than 14 inches in diameter. The wheels shall comply with ANSI B7.1. Grinding wheels for horizontal grinding shall be either straight (Type 1) or recessed (two sides,

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Type 7) complying with ANSI B74.2. If separate grinding wheels are necessary for grinding various metals (see 3.5), one wheel of each such type shall be provided with each machine.

3.6.4 Wheel head. The wheel head spindle shall rotate in bearings sealed against abrasive dust. Vertical feed shall be accomplished by micrometer type adjustment graduated in 0.001 inch increments. The feed adjustment shall be accessible to the operator.

3.6.5 Coolant system. A gear or impeller type pump shall be provided to force the coolant to the grinding wheel, to flush the wheel free of grinding residue, and to cool the part being ground. The capacity of the coolant system shall be of sufficient size to provide an adequate amount of coolant during continuous operation of the machine. Splash guards or splash rings shall direct the coolant back to the reservoir and prevent coolant loss. Provision shall be made to settle or filter the residue from the coolant before the coolant is returned to the reservoir. Means shall be provided for complete flushing and draining of the coolant reservoir.

3.6.6 Electrical equipment. Electrical equipment shall conform to NFPA 79.

3.6.6.1 Motors. The main drive motor shall be rated at not less than 7 1/2 horsepower, continuous duty, conforming to NEMA MG-1 with class B insulation. The main power shall be wired to operate from 230-460 volts, three phase, 60 hertz power source, and initially wired to operate from 230 volts. Feed and pump motors shall be induction, squirrel cage type, conforming to NEMA MG-1.

3.6.6.2 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.7 Standard equipment. The grinding machine shall be furnished with all equipment, work holding provisions, and attachments necessary to comply with the performance requirements specified in 3.5 through 3.5.2. Standard equipment shall include but shall not be limited to dresser, wheel, guard, wheel dresser, clamps, blocks, holding fixtures, and jaw fingers.

3.8 Spare parts. Unless otherwise specified (see 6.2.1), the following spare parts shall be furnished with each grinding machine:

(a) One extra belt of each type used.
(b) One extra grinding wheel of each type required (see 3.6.3).

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3.9 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent, i.e., epoxy enamel, shall be applied.

3.10 Product identification. The grinding machine shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

3.11 Workmanship. Standards of workmanship shall assure that the grinding machine shall have the stability, strength, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products submitted to the government for acceptance comply with all requirements of the contract. sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

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4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Accuracy test (see 4.6.1).
(c) Packaging inspection (see 4.7).

4.4.1 Sampling. Sampling for quality conformance inspection shall be performed as listed with all samples selected randomly.

Inspection or Test	Lot or Batch Size	Sample Size
Product examination (4.5)	1-90	100 %
Accuracy test (4.6.1)	1-90	13
Packaging inspectoin (4.7)	1-90	3

The lots shall not exceed the maximum sizes indicated above. If lot size is less than or equal to sample size, 100 percent inspection is required. Each lot shall be accepted with no defects and rejected if one or more defects are found.

4.5 Product examination. Visually, dimensionally, and manually examine each grinding machine to determine conformance with the requirements of 3.2 through 3.4.5 and 3.6 through 3.11. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the grinding machine passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.2.7), shall be cause for rejection.

4.6 Tests.

4.6.1 Accuracy test. The grinding machine shall be set up and operated in accordance with the manufacturer's instructions. A cylinder head from a standard size, commercial, six cylinder engine shall be ground to the extent

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necessary to refinish the entire surface. The ground surface shall be tested with precision equipment to determine conformance to the requirements (see 3.5.1).

4.6.2 Performance test. The grinding machine shall be operated in accordance with the manufacturer's instructions while using the standard equipment noted in 3.7. The wheel of the grinder shall first be dressed and the grinding machine shall then be used to completely refinish the gasket bearing surfaces of a six cylinder engine block, a V-8 engine block, the heads from those two engine blocks, and the intake and exhaust manifolds from both those engines. The grinding wheel shall be redressed as required. The refinished areas shall then be tested as specified in 4.6.1.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5 PACKAGING REQUIREMENTS

5.1 First article. When specified (see 6.2.1), the supplier shall furnish one complete pack for first article inspection. Whenever first article inspection is specified or performed, unit pack examination and tests for packaging may also by performed (see 3.10 and 4.2.1). The first article may be either a preproduction pack or an initial production pack which conforms to the requirements of this specification. In either case, the approved first article and the production packs shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

5.2 Disassembly. When practical, for the protection of components, attachments and accessories against damage, pilferage or to reduce cubage, items may be disassembled only to a necessary basic degree. Disassembled parts shall be preserved, anchored, braced, blocked, and cushioned to prevent damage. Disassembled parts shall be clearly and legibly marked as to identity and proper location on the assembled item. All fasteners removed during disassembly shall be secured in one of the mating parts. A part shall not be removed from an assembly unless it can be reassembled readily in the field without special skills or tools.

5.3 Disassembled parts marking. Disassembled parts shall be marked, when necessary with instructions, to facilitate reassembly. Removed parts and mating parts on the item shall be marked by stenciling, on the part, or by use of tags. Tags shall be waterproofed and ink used (on parts or tags) shall be waterproof. Tags shall be firmly secured to each individual part.

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5.4 Materials. Materials used in preservation-packaging and packing of machinery shall be as specified herein. Materials shall be new and free from all defects and imperfections which may affect the serviceability of the completed package or pack.

5.5 Processing conditions. Preservation-packaging shall be accomplished within buildings which have facilities to limit the accumulation of dust and moisture on the equipment being processed. Machinery brought into processing areas from outside shall be allowed sufficient time to reach ambient temperature before processing. Cleaning, drying, and application of preservatives shall progress in an uninterrupted series of operations. When periods of interruption are necessary, temporary protection shall be provided to partially processed items, as required, to insure against contamination.

5.6 General requirements. Component parts of machinery, accessories, and maintenance tools shall be cleaned, dried, preserved, packaged, and packed as specified herein. Parts and assemblies which are not covered by this specification shall be preserved and packaged utilizing the materials and methods specified for similar parts and assemblies described herein. Cleaning and drying procedures and preservatives and their application, specified herein, shall conform to the criteria and guidelines of MIL-P-116.

5.7 Preservation-packaging. Preservation-packaging shall be Level A or C, as specified in the contract (see 6.2.1).

5.7.1 Level A.

5.7.1.1 Preservation-packaging. Unless otherwise specified (see 5.7.1.25), the basic machine shall be preserved and packaged in accordance with method I criterion of MIL-P-116. Electronic and electrical equipment, hand tools and accessories shall be preserved and packaged as specified herein. When it has been determined that a preservative is required to protect a specific surface or item, but has not been specified herein, the preservative utilized shall be selected from MIL-P-116.

5.7.1.2 Cleaning and drying. Thorough cleaning and drying of all bare ferrous metal surfaces is essential before the application of a preservative or other protective finish such as paint, plating, etc.. Care must be taken to limit the handling of cleaned parts prior to the assembly of new machinery. Where practicable, processors and assemblers shall wear rubber or plastic gloves while handling cleaned parts. Unless otherwise specified herein, machines which have been functionally tested prior to shipment shall have all operational systems thoroughly drained before the application of a preservative. It may be necessary to connect the machine to a power source and operate at low speeds in order to drain all cavities. Extreme care shall be taken to immediately disconnect power when machine is being drained so as not to damage pumps, motor or other components.

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5.7.1.3 Internal surfaces. All ferrous metal surfaces of parts which will be utilized internally in the assembly of new machinery shall be cleaned in accordance with process C-5 and dried in accordance with the applicable procedure of MIL-P-116. In the interim, to preclude contamination from forming on cleaned surfaces prior to the assembly operation, parts shall be coated with a light weight preservative oil conforming to P-9. To prevent dust from accumulating on cleaned parts, cover with a sheet of greaseproof material, such as MIL-B-121. In the event the supplier deems it necessary to clean a particular mechanism after draining the operational lubricant, flush with a solution of one part lubricating oil P-10, type I, grade 30 and nine parts cleaning solvent conforming to P-D-680. Otherwise, the mechanisms need not be cleaned. All ferrous metal internal surfaces shall be coated with the lubricant or a preservative. Internal mechanism shall be cycled to assure all areas are protected with a coating of the lubricant before either sealing or draining. Any unprotected areas will have a preservative coating applied.

5.7.1.4 External surfaces. All accessories and attachments such as chucks, faceplates, steady rests, toolholders, and abrasive wheels shall be removed from the assembly prior to the external cleaning operation. All external, critical ferrous metal surfaces (fit or function) of the machine shall be cleaned in accordance with process C-5 and dried in accordance with the applicable procedures of MIL-P-116. Noncritical ferrous metal surfaces shall be cleaned in accordance with process C-3 and dried as specified above. Nonmetallic composites and painted metal surfaces shall be cleaned by wiping with a clean, dry cloth. (Care must be taken to prevent cleaning solvents from coming in contact with nonmetallic composites). When cleaned surfaces of the machine show signs of rusting before application of the final preservative, the cleaning process shall be confined to those specific small areas of the assembly. Alternatively, the solvent used for the final wiping operation shall contain a small amount (approximately 5 percent by volume) preservative oil conforming to P-10, type 1, grade 30. (It is extremely important that elapsed time between these operations be held to an absolute minimum).

5.7.1.5 Steam and air lines. Steam and air lines shall be drained of all water accumulation and dried by injecting moisture-free air into the systems.

5.7.1.6 Disassembled components. Parts detached from the basic machine shall be cleaned and dried as specified in 5.7.1.2 and 5.7.1.3.

5.7.1.7 Maintenance or machine tools and tool accessories. All machine, maintenance, metal or woodworking, hand tools and any accessories there of shall be preserved and packaged in accordance with Level A requirements of PPP-P-40.

5.7.1.8 Electrical equipment. Electrical equipment, whether attached or detached, shall be cleaned, dried and preserved in accordance with Level A requirements of MIL-E-17555 or MIL-E-16298, which ever applies to the individual electrical item.

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5.7.1.8.1 Electrical motors. Electric motors, whether attached or detached, shall be preserved and packaged in accordance with Level A requirements of MIL-E-16298, and applicable subparagraph for assembled machines.

5.7.1.8.2 Electrical control devices. Control units, whether attached or detached, shall be preserved and packaged in accordance with Level A requirements of MIL-E-17555.

5.7.1.8.3 Electrical wiring system. Exposed ends of wires, plug openings, sockets, connector plugs, terminals and openings into switches and junction boxes shall be sealed with tape or cap-plugs conforming to PPP-T-60, type IV or MIL-C-5501, respectively. Electric cable shall be coiled to a minimum safe diameter and secured with cotton twine or tape to prevent uncoiling.

5.7.1.9 Hoses and Fittings. Hoses and fittings which are not attached to the basic machine shall be cleaned, dried and preserved in accordance with Level A requirements of MIL-P-775.

5.7.1.10 Lubricating systems. The machine, with all lubricants, greases and other lubricating compounds either installed or included (see 3.8.2.9) shall have all vent openings in the lubrication systems temporarily capped, plugged or taped closed to prevent loss of lubricants during shipment and storage. Tags, warning that "ALL TEMPORARY CAPS, PLUGS OR TAPE MUST BE REMOVED BEFORE START-UP OR USE OF THE ITEM", shall be attached to all temporary caps, plugs or tape intended to be vented to the atmosphere during operation. In the event a lubrication reservoir cannot be covered and sealed, it shall appropriately be preserved and the lubricant shall be provided in a sealed container and shipped with the machine (see 3.8.2.9). A warning tag shall be placed in a conspicuous place near the filling area, warning the user to "INSTALL THE FURNISHED LUBRICANT BEFORE OPERATION OF THE EQUIPMENT".

5.7.1.11 Coolant systems. Coolant systems shall be drained and cleaned of all coolant residue and shall be preserved by flushing with P-3 preservative compound leaving a residual film of the preservative in the coolant lines and tank. Drain off the preservative and close the systems. Interior surfaces of the tank/reservoir which have not been covered or flooded with preservative while the machine is in operation shall be manually coated by brushing or spraying with similar type preservative.

5.7.1.12 High-speed spindle bearings. Reservoir shall be filled to normal operating level with manufacturer's lubricant and sealed closed for shipment and storage. If the system is of such design that the lubricant will leak out during shipment and storage, it shall be drained. The system shall then be preserved by filling with preservative oil P-10, type I, grade 30, drained, sealed and covered.

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5.7.1.13 Hydraulic systems. Both open- and closed systems shall be filled to normal operating level with either the manufacturer's operative hydraulic oil or hydraulic oil conforming to MIL-H-6083. After filling, the machine shall be cycled to circulate the oil through the entire system. Prior to disconnecting the machine from the power source, place the piston(s), except opposing pistons, in their innermost position. The oil shall remain in the reservoir, providing the reservoir can be sealed to prevent any leakage during shipment and storage. If it cannot be sealed to prevent leaking, it shall be drained and preserved and the hydraulic oil shall be provided in a sealed container and shipped with each machine.

5.7.1.14 Grease fittings. Charge all grease fittings with manufacturer's compound or with preservative compound conforming to P-11.

5.7.1.15 Nonlubricated interior surfaces. Surfaces of internal gears which do not run in lubricant shall be thoroughly coated with P-7 preservative.

5.7.1.16 Gear cases. Fill the gear cases with normal lubricating oil. Shift gears into all possible positions while operating the machine. Make sure all surfaces are coated with lubricating oil and close and seal all openings to prevent any leakage during shipment and storage.

5.7.1.17 Sliding of friction surfaces. Surfaces of ways, gibs, driving gears, traversing and adjusting screws, and other critical, exterior operating machined surfaces shall be coated with P-2 preservative compound. Parts of the machine which are adjustable shall be locked into place by means provided and blocked to assure movement shall be kept to a minimum.

5.7.1.18 Bearings and journals. Grease lubricated bearings shall be preserved with type P-11 preservative. Oil lubricated journals shall be filled with normal oil lubricant or with lubricating oil P-10, type I, grade optional.

5.7.1.19 Drive belts and pulleys. Drive belts shall either be removed from the machine or released from tension. Belts removed from the machine shall be packaged in accordance with method IC-1 or IC-3 criterion of MIL-P-116. Ferrous metal pulleys shall be coated with a rust-inhibiting primer conforming to TT-P-664.

5.7.1.20 Grinding wheels and abrasives. All abrasive products shall be preserved and packaged in accordance with Level A requirements of MIL-P-3816.

5.7.1.21 Closed dial indicators. Closed dial indicators shall be coated with 1 inch thick uncompressed cushioning materials conforming to PPP-C-843, type II, class B. Secure cushioning material in place with tape conforming to PPP-T-60, type IV. Other type cushioning materials and tapes may be used provided the materials are water-resistant, non-corrosive, and will provide equivalent or better protection of those specified above.

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5.7.1.22 Drawers and door assemblies. Sliding surfaces of drawer guides shall be coated with P-11 preservative grease. Interior surfaces of door hinges shall be lubricated by the application of P-10, grade 30 preservative oil. In addition to mechanical locks or catches incorporated in the machine tool, secure doors and drawers with metal strapping or other devices which are suitable for the purpose intended. Fiberboard pads shall be placed under metal strapping, to prevent marring the painted surfaces.

5.7.1.23 Closure of openings. Small openings (except vents and louvers installed for ventilation purposes) which will admit dust or water shall be sealed with tape conforming to PPP-T-60, type IV. Large openings shall be covered with waterproof barrier material conforming to class B-1 or L-4 of PPP-B-1055, or, as a minimum, plastic material conforming to L-P-378, type and class optional. Secure materials in place with tape specified above. When very large openings are covered, or when locations of openings render covering vulnerable to puncture, coverings shall be protected by applying wood, plywood or sheet metal over the materials. Open ends of all piping and fittings shall be plugged to prevent the entrance of foreign materials, using pipe fittings, plastic cap-plugs conforming to MIL-C-5501 or tape conforming to above.

5.7.1.24 Technical data. Technical data furnished with the machine shall be packaged in accordance with method IC-1 criterion of MIL-P-116. The package shall be marked "TECHNICAL DATA" in bold black letters. The package shall be protected against puncture and abrasion. The data shall be either attached to the machine with tape conforming to PPP-T-60, type IV, placed in a stowage compartment of the machine, or positioned in a consolidated box containing accessories or disassembled components.

5.7.1.25 Method II. When specified (see 6.2.1), complex items which cannot be treated entirely externally with a contact preservative or where removal of the preservative would prevent timely use of the machine shall be packaged in accordance with method II criterion of MIL-P-116. Accessories, maintenance tools, control units, and all internal surfaces of the machine shall be preserved and packaged as specified herein. Method II packs will bear precautionary labels as stated in MIL-STD-129. Upon receipt of a Method II package, it shall be inspected to assure the vacuum seal is intact before being placed in storage or shipped elsewhere.

5.7.1.26 Protection of preservatives. All exposed preserved surfaces of machines which are subject to being disturbed during the packing operation shall be loosely covered with barrier material conforming to MIL-B-121, type 1, grade A. Not less than two layers of similar-type barrier material shall be used to separate wood or metal blocking and finished metal surfaces of the machine. When considered necessary, secure barrier in place with tape conforming to PPP-T-60, type IV.

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5.7.1.27 Consolidated packaging. When size and weight permit, packaged accessories, maintenance tools, and small disassembled components shall be consolidated and placed in drawers and stowage compartments of the machine. When it is not feasible to stow parts as specified above, place parts in box(es) conforming to PPP-B-636, class WR or WWVR; PPP-B-640, class 2; PPP-B-601, domestic type; or PPP-B-621, class 1. Large bulky disassembled components shall be packed as specified in 3.7, as applicable.

5.7.1.28 Cushioning. Cushioning materials and methods of application shall conform to the criteria and guidelines of MIL-P-116 and MIL-STD-1186. In the event there are conflicting requirements, MIL-STD-1186 will have precedence.

5.7.1.29 Movable parts. Operating heads, sliding tables, rams, counter-balances, and other movable parts shall be secured by mechanical devices incorporated in the machine. The devices alone do not always provide adequate support to prevent movement in the handling and transporting of the item and therefore must be augmented with appropriate wood blocking, metal tie rods used in conjunction with wood blocking, metal strapping, or other devices which will immobilize the part. Handwheels, cranks, levers, and similar controls shall be secured in position with common annealed wire or steel strapping.

5.7.1.30 Positioning assemblies. Assemblies having vertical movement shall be blocked and braced in a lowered position; those capable of moving traversely shall be centered for blocking and bracing.

5.7.1.31 Tables. Tables and other components moving on ball bearings or other types of high efficiency, low-friction ball or roller bearing assemblies shall be removed, or blocked, and all components treated in a such a manner that neither the way(s), surface(s), nor the anti-friction devices will be subject to brinelling or other damage. Measurement, indicating, etc., shall be individually enclosed with not less than 1 inch uncompressed cushioning and placed in a snug-fitting fiberboard box. To provide maximum protection the fiberboard box(s) shall be placed in a wooden box conforming to PPP-B-601, domestic, style optional, for later consolidation into the shipping container.

5.7.2 Level C preservation. Preservation of machinery shall conform to the requirements of MIL-STD-1190.

5.8 Packing. Packing shall be Level A, B or C, as specified in the contract.

5.8.1 General requirements for Levels A and B. When weight, size and design permit, machines shall be assembled when being prepared for shipment. When it is not considered feasible to ship a machine assembled, components shall be packed in accordance with weight limitations specified herein.

5.8.2 Level A.

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5.8.2.1 Machinery weighing less than 2,500 pounds. Unless otherwise specified (see 6.2.1), machinery including accessories, weighing less than 2,500 pounds, shall be packed in cleated plywood boxes conforming to PPP-B-601, overseas type; nailed wood boxes conforming to PPP-B-621, class 2, style 2, 3, 4 or 7; or skidded boxes conforming to MIL-B-26195, type II, style A, class optional, with plywood superstructure. Selection of containers shall be governed by the size and weight limitations of the applicable container specification. Machines designed with mounting holes in areas that can withstand rough handling without breakage shall be bolted to an auxiliary base before being packed in a box conforming to PPP-B-601 or PPP-B-621. Contents of boxes shall be arranged, anchored, blocked, braced and cushioned in accordance with MIL-P-116 and MIL-STD-1186. In the event there are conflicting requirements, MIL-STD-1186 will have precedence. Top-heavy machines shall be packed as specified in 3.8.2. . Box closure and strapping requirements shall be in accordance with the appendix to the applicable box specification.

5.8.2.2 Machinery weighing 2,500 to 30,000 pounds. Unless otherwise specified (see 6.2.1), machines weighing 2,500 to 30,000 pounds shall be packed individually in crates conforming to MIL-C-104, type I, class 2, style A. To conserve cubage, safeguard components of the machine, and to comply with weight and dimensional limitations of the crate specification, subassemblies removed through disassembly and control units may be packed in separate shipping containers. When packed separately, items weighing less than 2,500 pounds shall be packed as specified in 3.8.2.1. Arrangement of contents, anchoring, blocking and cushioning shall be in accordance with the appendix to the crate specification, MIL-STD-1186, and as specified herein. MIL-STD-1186 will have precedence when there are conflicting requirements. When practicable, to facilitate loading into common carriers, the height of the crated machine shall be limited to 9 feet 8 inches.

5.8.2.3 Buttress blocking. Hold down bolts on machines weighing over 8,000 pounds shall be reinforced with backup blocking extending from the machine base to the skid header; blocking shall be parallel to skid runners and in line with bolt holes on the machine. The blocks shall be similar to that used for the skid header. Carriage or step bolts shall be utilized to secure the block to the skid base.

5.8.2.4 Unsupported columns. Unsupported columns of the machine shall be reinforced by placing snug-fitting blocks between the columns and the skid platform or machine base, as applicable. Blocks extending from the column to the skid base shall be fitted so that tightening of the hold down bolts will not place undue stress on the column(s).

5.8.2.5 Top-heavy machinery. To safeguard equipment and personnel, top-heavy machine tools may be packed in a horizontal position, design permitting. Blocking and bracing shall be located at points of the machine with sufficient rigidity to carry the machine weight. Crates for top-heavy

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items requiring skidding due to design, structural weakness, etc., shall be sized to provide stability in handling and transit. The length and width of the skid platforms shall extend beyond the machine a distance equivalent to the height of the center of gravity of the machine above the base.

5.8.2.6 Leg-type machines. Leg-type machines with slotting holes for mounting shall be reinforced by placing a nominal 2- by 4-inch wood block against the leg opposite the slot opening and nailing to the skid base.

5.8.2.7 Bolting. Machines shall be secured to the skid base by not less than four bolts. When sufficient bolting points are not available, J-clamps or tie rod assemblies as specified in 3.7.1, or other devices, shall be used to provide the minimum of four holding points. Bolts shall pass completely through the skid runner or sleeper, as applicable, and base flooring.

5.8.2.8 Consolidated containers. Consolidated containers specified in 3.7.1.25 shall be placed inside the shipping container housing the machine. The boxes shall be restrained by the use of metal strapping of sufficient strength or tie rod assemblies.

5.8.2.9 Operational oils. Operational oils furnished with the machine, but not placed in the machine stowage receptacles, shall be stowed in metal drum(s) conforming to PPP-D-705, type I or II or PPP-D-729, type I or II. Drums shall be positioned upright inside the shipping container housing the machine. Sufficient blocking shall be applied to immobilize the drum(s) inside the shipping container. Alternatively, when the drum(s) tend to increase the cubage of the shipping container, the drum(s) may be shipped as individual containers.

5.8.2.10 Machines exceeding crate limitations. Unless otherwise specified (see 6.2.1), machines exceeding the weight limitations if MIL-C-104 shall be packed in a similarly designed crate. The crate's structural members and anchoring devices shall be increased in number and size as needed to accommodate the load to be packed therein.

5.8.3 Level B. Machines weighing 16,000 pounds or less shall be packed for shipment utilizing boxes conforming to PPP-B-601, domestic type, style A, B, I, or J; PPP-B-621, class 1, style 2, 3, 4, or 7; MIL-B-26195, type I, style A; or crates conforming to MIL-C-52950 or MIL-C-3774. Selection of the container shall be governed by the size and weight limitations of the applicable container specification. Machines designed with mounting holes in areas that can withstand rough handling without breakage shall be bolted to an auxiliary base before being packed in a box conforming to PPP-B-601 or PPP-B-621. Arrangement of contents, anchoring, cushioning, blocking, bracing and container closure shall be the same as for level A (see 3.8.2.1). Items packed in open crates shall be covered with a waterproof shroud conforming to MIL-STD-1186. Machines exceeding 16,000 pounds shall be packed as specified in 3.8.2.2.

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5.8.4 Level C packing. Packing of metal and woodworking machinery shall conform to the requirements of MIL-STD-1190. When authorized by the transportation media and when safe handling with mechanical equipment is provided, the machines may be mounted on skids of a skid base in lieu of being boxed or crated. Openings in skids shall be a minimum of 3 inches in height to provide access for forklift tines. Machines not packed in fully sheathed containers shall be completely covered with a waterproof plastic shroud. The shroud shall be secured to the base of the machine. Polyvinyl chloride (PVC) and ethylene vinyl acetate (EVA) shall not be acceptable as an intimate wrap or shroud material due to the possible corrosive effect of PVC and EVA vapors on the shrouded item and its component parts.

5.9 Marking. All marking shall be in accordance with MIL-STD-129. Bar code markings are required and shall be in accordance with MIL-STD-129. Center of balance markings and arrow markings, indicating THIS SIDE UP, shall be applied to each shipping container for all levels for the protection of the machinery and personnel.

5.10 Rough handling. The completed unit pack, shipping container(s) and all shall be capable of withstanding the shipping and handing conditions expected from fork lift handling and truck transport. The container, its contents and the packaging shall not be damaged, functionally or physically, or items displaced by the appropriate rough handling test(s) specified elsewhere in the contract.

6. NOTES

6.1 Intended use. The grinding machine covered by this specification is intended to be used primarily to recondition the gasket bearing surfaces of cylinder heads, engine blocks, manifolds, and other similar flat surfaces.

6.2 Ordering data. Procurement requirements in 6.2.1 which must be exercised and required of the supplier contractor shall be specified in the solicitation, contract or purchase agreement.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Measuring and indicating device graduation(s) required (see 3.2.2).
(d) Exceptions and additional safety/health requirements when required (see 3.2.8 and 6.4).
(e) Spare parts if different (see 3.8).
(f) Painting and finishing if different (see 3.9).
(g) Product identification if different (see 3.10).
(h) First article inspection if different (see 4.3).

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(i) Quality conformance inspection if different (see 4.4).
(j) Specify level of preservation/packaging and packing required (see 5.7 and 5.8).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract (see 6.3.1).

6.3.1 Repair part manuals. Repair part identification, in repair part manuals, shall include the prime contractor's name, nomenclature, part number, and price. Repair part identification shall also include the name of the original equipment/part manufacturer (OEM - actual part manufacturer), Commercial and Government Entity Code (CAGEC), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be so identified as shown in MIL-M-7298.

6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.5 Subject term (key word) listing.

AMSC number
Design
First article
First article inspection
grinding machine
grinding wheel
horizontal grinding
Materials
Ordering data
Performance
Protective finish
Scope
vertical grinding
wet grinding aluminum
wheel head
wheel spindle

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Army - AL
Air Force - 99

Review activities:

Navy - MC
Air Force - 82

Preparing activity:

Army - AL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-G-45851E	03 Oct 1991

3. DOCUMENT TITLE

Grinding Machine, Cylinder Head (Wet Type)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
[redacted]	
c. ADDRESS (Include Zip Code) [redacted]	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

7. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) DSN
Cmdr USA Armament Res, Dev. & Engr Ctr	(319) 782-8230	793-8230
c. ADDRESS (Include Zip Code) ATTN: SMCAR-EST-S Rock Island, IL 61299-7300	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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